

BIG DATA IMDB MOVIE REVIEW ENGINE

Final Year Project 2024
Supervisor: **Dr. Farhan Qamar**

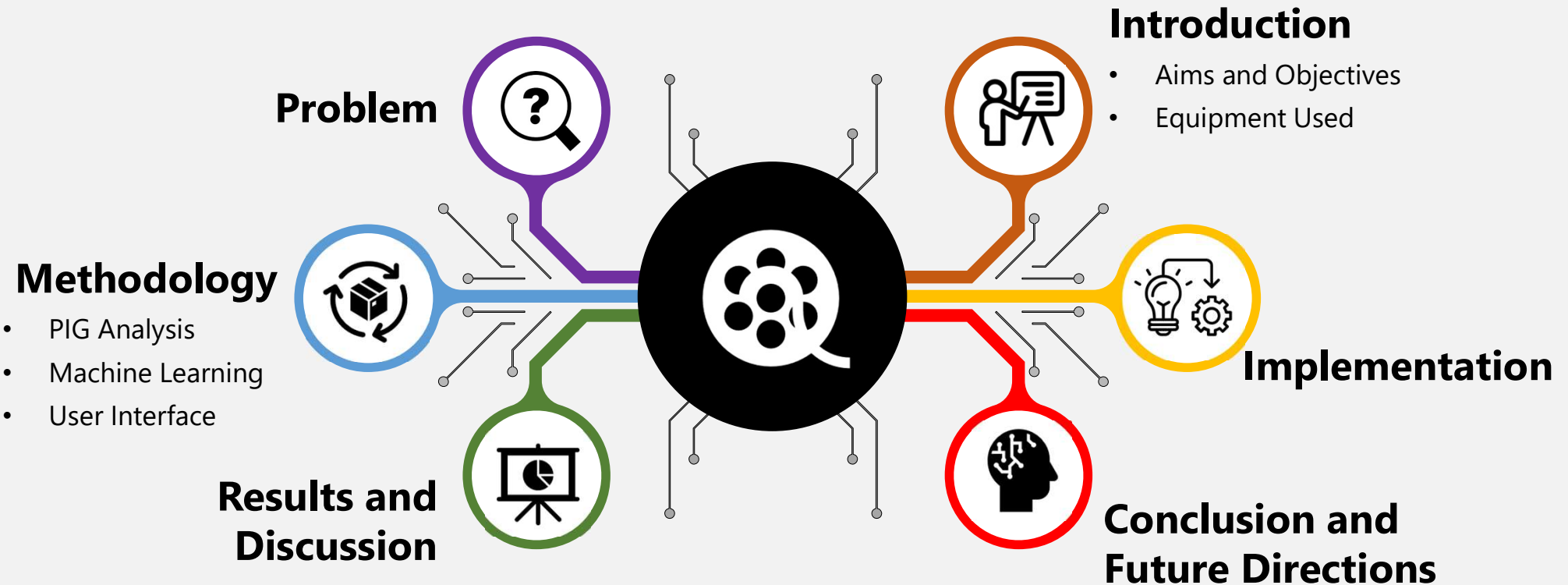
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SHAHER BANO SANI (20-CP-50)



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The current recommendation and review systems faces limitations in **storage**, **processing**, and **analysis** of large amounts of data which leads to issues such as poor data quality, limited ability to handle increasing demands, lack of personalized recommendations, and inadequate labeling of reviews.



BIG DATA BASED MOVIE RECOMMENDATION SYSTEM AND REVIEW ENGINE



Big Data
Analysis



Content-
based
suggestions
based on
movie
metadata



Sentiment
Analysis
based on user
feedback



Machine
learning
integrated
front-end



AIMS

1. Big data-based IMDB movie review engine
2. Utilization of different big data platforms like Apache Hive and Pig for efficiently managing large movie datasets



OBJECTIVES

1. Collection of data from IMDB
2. Analysis of data using pig Latin
3. ML algorithm for sentiment analysis
4. Content-based recommendation system
5. User-friendly interface



SUSTAINABILITY DEVELOPMENT GOALS



Goal 9: Industry, Innovation, and Infrastructure



TOOLS

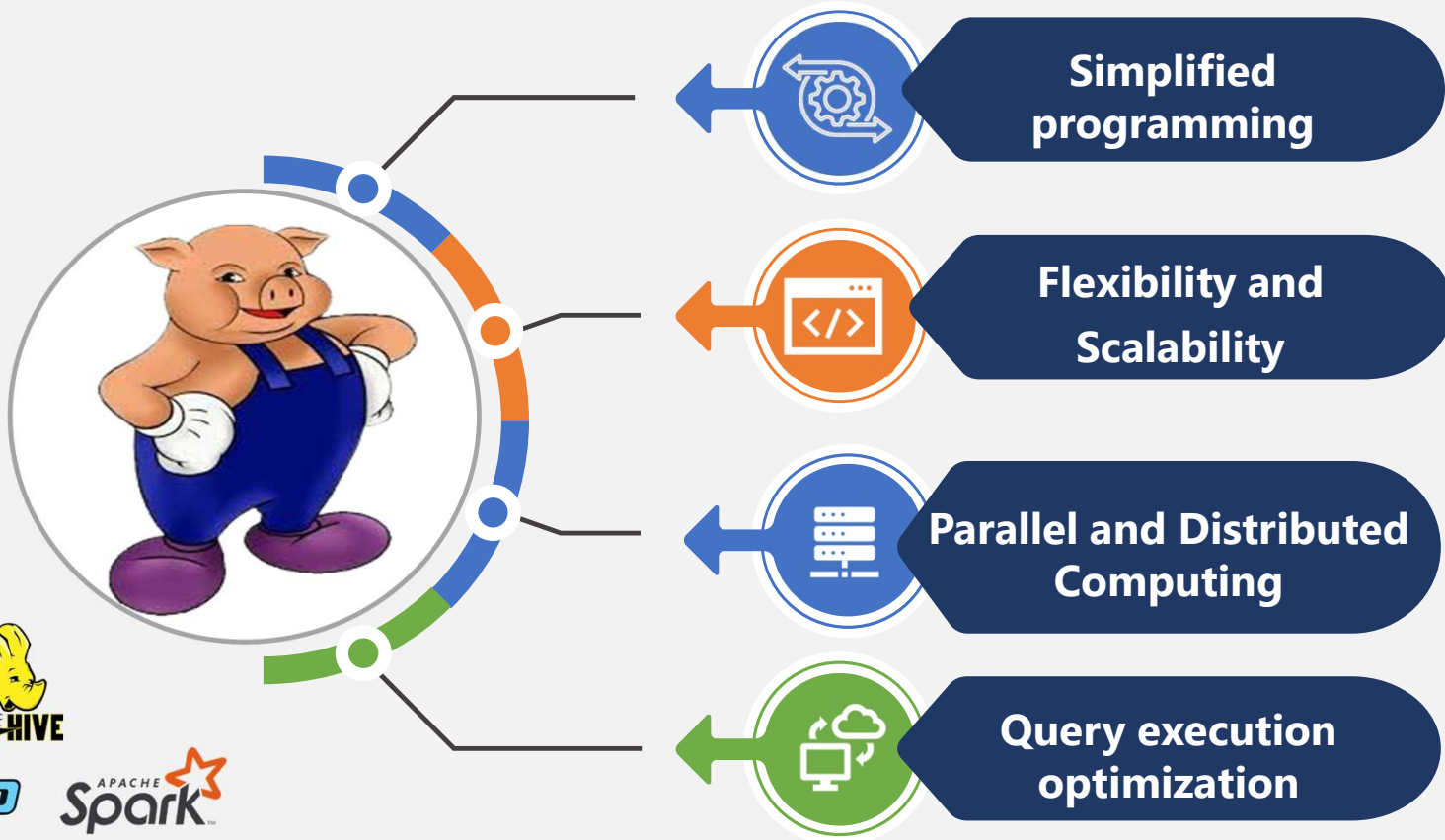
- Apache Pig
 - Pig Latin
- Hadoop Distributed File System (HDFS)
- Jupyter Notebook
- VS Code

LIBRARIES

- Piggybank
- Scikit-learn
 - TFIDF Vectorizer
 - Cosine Similarity
- NLTK (Natural Language Toolkit)



WHY APACHE PIG?

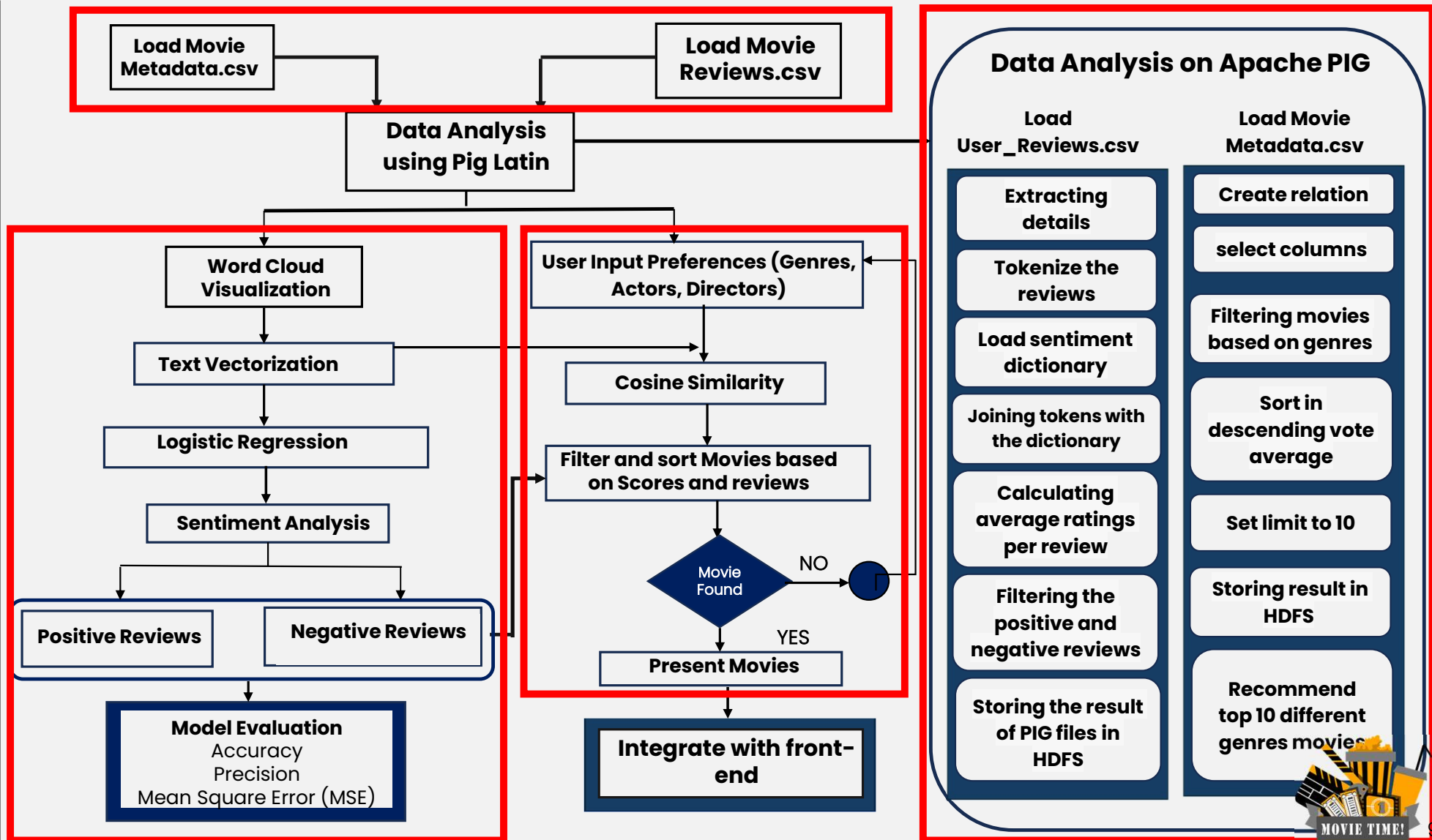


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METHODOLOGY





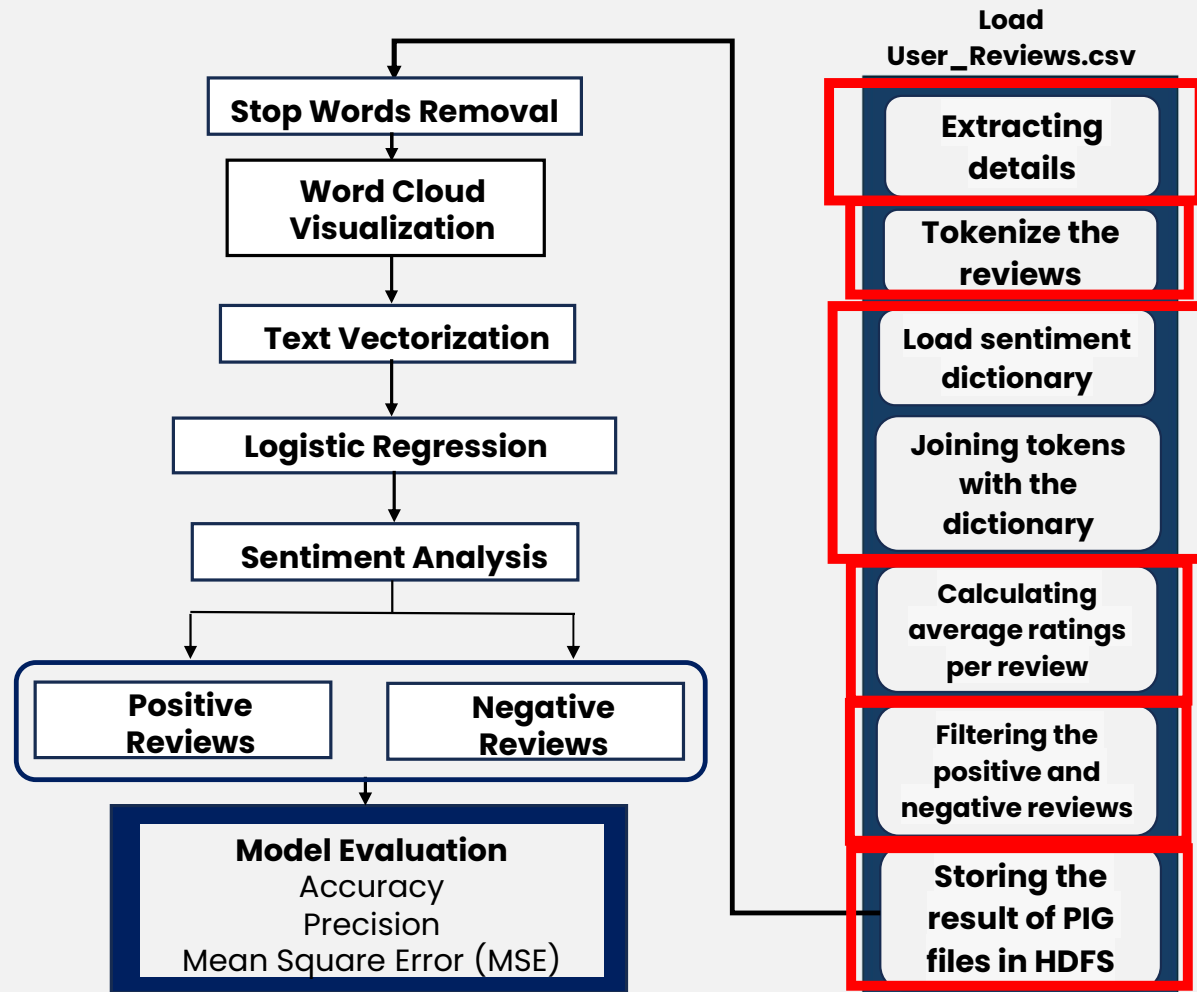
DATA COLLECTION

- Dataset Source
- Why IMDb?

**MOVIE
METADATA**

**MOVIE
REVIEW**



**ANALYSIS OF MOVIE REVIEWS DATA**

NEGATIVE WORD CLOUD VISUALIZATION



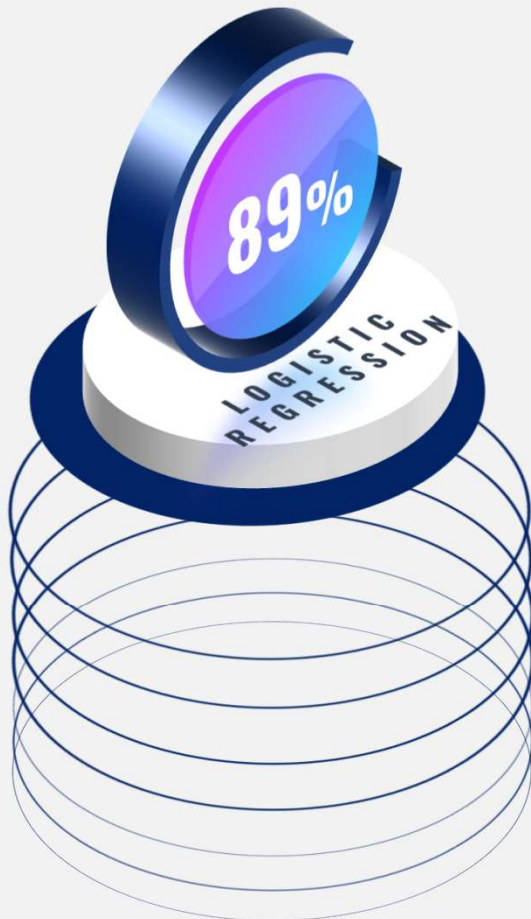
COMPARISON OF MODELS

MODEL NAME	ACCURACY	PRECISION	RECALL	F1 SCORE
Logistic Regression	0.89	0.89	0.89	0.89
Random Forest Classifier	0.86	0.86	0.84	0.85
Naïve Bayes	0.85	0.85	0.85	0.85



COMPARISON OF MODELS

LOGISTIC REGRESSION



RANDOM FOREST



NAÏVE BAYES



Home

Problem

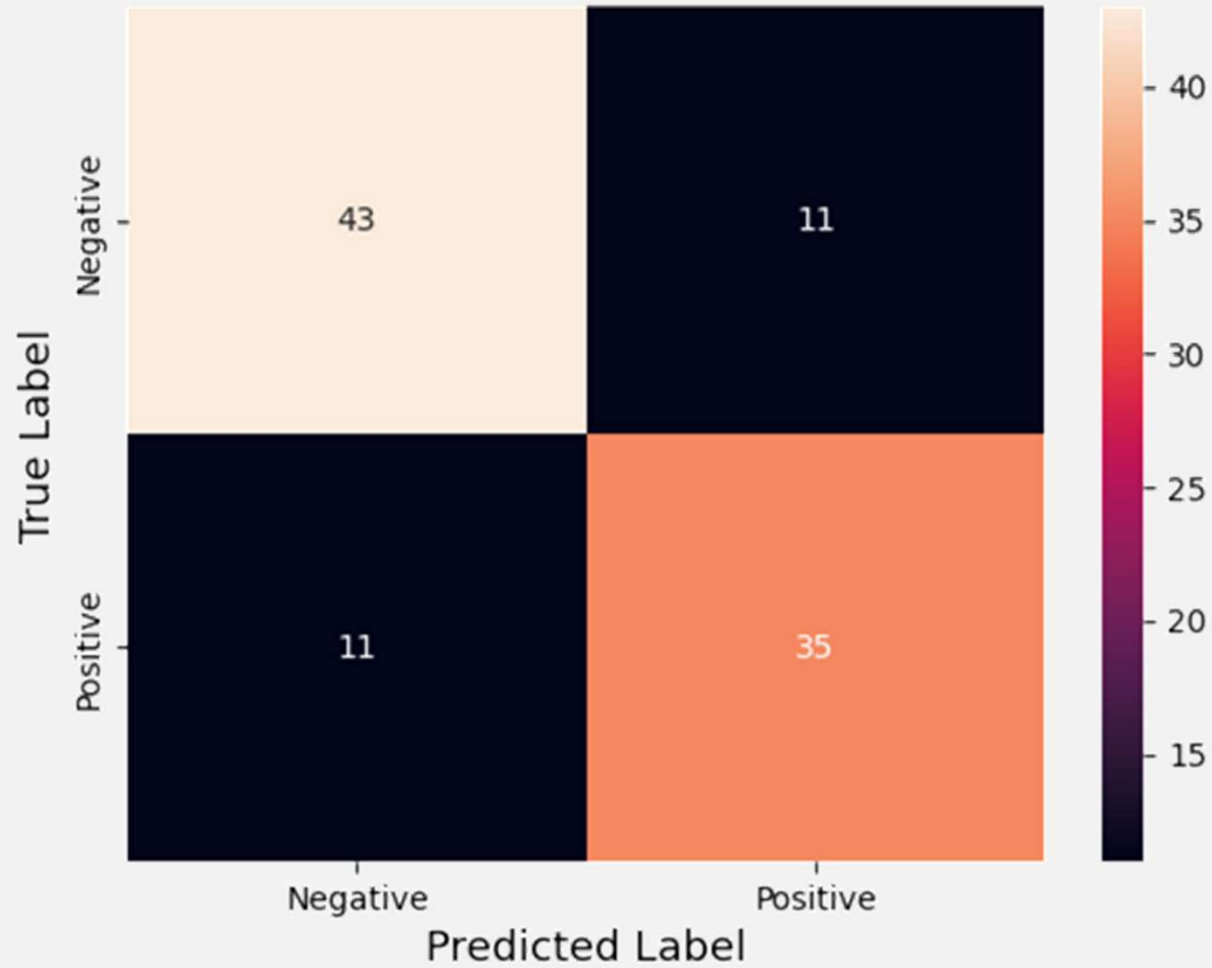
Introduction

Methodology

Results and Discussions

Conclusion

Future Directions



ANALYSIS OF MOVIE METADATA

Load Movie
Metadata.csv

Create relation

select columns

Filtering
movies based
on genres

Sort in
descending
vote average

Set limit to 10

Recommend
top 10 different
genres movies

Storing result in
HDFS

User Input Preferences
(Genres, Actors,
Directors)

Text Vectorization

Cosine Similarity

Filter and sort Movies
based on Scores and
reviews

MOVIE
FOUND

NO

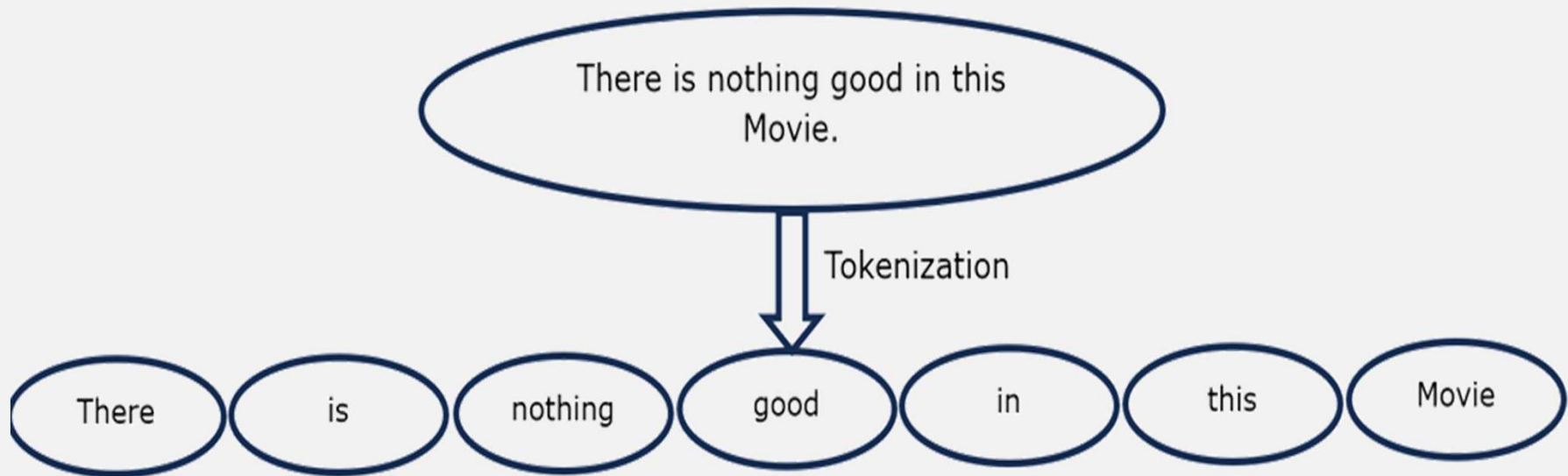
YES

Present Movies

Integrate with front-
end



TOKENIZATION



POSITIVE REVIEWS

Movie_Reviews - COMPLETED

Job ID job_1713449702041_0098

Started 2024-04-18 20:43

▼ Results

[Download](#)

((33542,I will not comment on the story as such. I agree with most peoples comments that this is a good all round acti
((33667,"I caught this on the dish last night. I liked the movie. I traveled to Russia 3 different times (adopting our
((33909,There are no reasons to watch this movie. Should you have won and extreme amount of money and having spent you

NEGATIVE REVIEWS

Movie_Reviews - COMPLETED

Job ID job_1713449702041_0101

Started 2024-04-18 20:45

▼ Results

[Download](#)

((32823,Once again Elmer is faced with the dilemma of who to shoot. Bugs of Daffy. He's unsure of what season it is an
((33155,This movie was sweet. The main character lady was sensitive to 2 different men who wanted her. She seemed not
((33196,This film is the worst film I have ever seen. The story line is weak - I couldn't even follow it. The acting i
((33644,Just watched this film on TV and it was awesome.

had just planned on watching it whilst doing some



← → ↻ ⚠ Not secure 192.168.48.152:8080/#/main/views/FILES/1.0.0/Files ☆ ⬇ 📄 👤 ⋮

Gmail YouTube Maps | 📁 All Bookmarks

Ambari Sandbox 0 ops 0 alerts Dashboard Services Hosts Alerts Admin admin ▾

/ user / admin + New directory Browse... Select files to upload. X

admin Search File Names 🔍

Name	Size	Last Modified	Owner	Group	Permission	Asc	Owner ▾	
..								
.staging	-	2024-02-03 19:19	admin	hdfs	-rwx-----			➡ 🗑️ ☐
CosineSimilarity.jar	3.5 kB	2024-02-03 18:14	admin	hdfs	-rw-r--r--			➡ 🗑️ ☐
WordCount-5.1.2.jar	1.6 MB	2024-02-03 18:13	admin	hdfs	-rw-r--r--			➡ 🗑️ ☐
commons-lang-2.6.jar	277.6 kB	2024-02-03 18:12	admin	hdfs	-rw-r--r--			➡ 🗑️ ☐
concat-0.1.14.jar	2.9 kB	2024-02-03 18:14	admin	hdfs	-rw-r--r--			➡ 🗑️ ☐
guava-13.0.1.jar	1.8 MB	2024-02-03 18:13	admin	hdfs	-rw-r--r--			➡ 🗑️ ☐
negative_reviewss	-	2024-02-03 19:18	admin	hdfs	-rwxr-xr-x			➡ 🗑️ ☐
positive_reviewss	-	2024-02-03 19:18	admin	hdfs	-rwxr-xr-x			➡ 🗑️ ☐



Welcome to the Movie Review Sentiment Predictor!

Please enter your review about the movie:

> This movie is really good, performances are great.

Welcome to the Movie Review Sentiment Predictor!

Please enter your review about the movie:

> This movie is really good, performances are great.

The sentiment of the review is positive



>

WELCOME!!! to the Movie Review Sentiment Predictor!
Please give your review about the movie:

Welcome to the Movie Review Sentiment Predictor!

Please enter your review about the movie:

> There is nothing good in this Movie.

The sentiment of the review is negative



Movie_Recommendation - COMPLETED

Job ID job_1713449702041_0034

Started 2024-04-18 19:46

▼ Results

(Big Hero 6,Adventure Family Animation Action Comedy,6.5210547E8)
(Teenage Mutant Ninja Turtles,Science Fiction Action Adventure Fantasy Comedy,4.772E8)
(Home Alone,Comedy Family,4.76684672E8)
(Pretty Woman,Romance Comedy,4.63E8)
(Happy Feet,Animation Comedy,3.84335616E8)
(Hotel Transylvania,Animation Comedy Family Fantasy,3.58375616E8)
(The Flintstones,Fantasy Comedy Family,3.416312E8)
(Chicken Little,Animation Family Comedy,3.14432672E8)
(Chicago,Action Comedy Crime Drama Music,3.06776736E8)
(Neighbors,Comedy,2.68157408E8)

[Download](#)

TOP 10 COMEDY
MOVIES



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BACK



Not secure 192.168.48.164:8080/#/main/views/FILES/1.0.0/Files

Ambari Sandbox 0 ops 0 alerts Dashboard Services Hosts Alerts Admin admin

/ user / admin + New directory Browse... Select files to upload.

admin

Search File Names

Name	Size	Last Modified	Owner	Group	Permission	Asc	Name	
..								
.staging	-	2024-03-02 21:45	admin	hdfs	-rwx-----			
TopActionMovies	-	2024-03-02 21:45	admin	hdfs	-rwxr-xr-x			
topAdventureMoviesWithTitle	-	2024-03-02 21:45	admin	hdfs	-rwxr-xr-x			
topComedyMoviesWithTitle	-	2024-03-02 21:45	admin	hdfs	-rwxr-xr-x			
TopCrimeMovies	-	2024-03-02 21:45	admin	hdfs	-rwxr-xr-x			
TopDramaMovies	-	2024-03-02 21:45	admin	hdfs	-rwxr-xr-x			
TopHorrorMovies	-	2024-03-02 21:45	admin	hdfs	-rwxr-xr-x			
TopThrillerMysteryMovies	-	2024-03-02 21:45	admin	hdfs	-rwxr-xr-x			



HYY!! WELCOME TO RECOMMENDATION SYSTEM

Enter the name of the Movie you want to watch:

MOVIE SUGGESTED FOR YOU ARE :

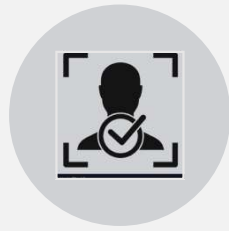
- 1 . The Avengers
- 2 . Avengers: Age of Ultron
- 3 . Captain America: The Winter Soldier
- 4 . Captain America: Civil War
- 5 . Iron Man 2
- 6 . Thor: The Dark World
- 7 . X-Men
- 8 . The Incredible Hulk
- 9 . X-Men: Apocalypse
- 10 . Ant-Man
- 11 . Thor



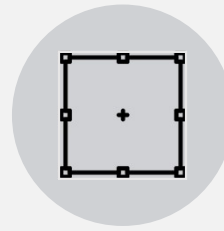
ACTOR-BASED RECOMMENDATION SYSTEM



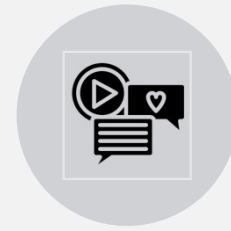
**Facial
Features
Extractions**



**Target Actor
Face
Detection**



**Creating
Bounding Boxes
For Identification**



**Matching Results
and
Recommendations**



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DATASET



26

IMPLEMENTATION

HYY!! WELCOME TO RECOMMENDATION SYSTEM

Enter the name of the Movie you want to watch:

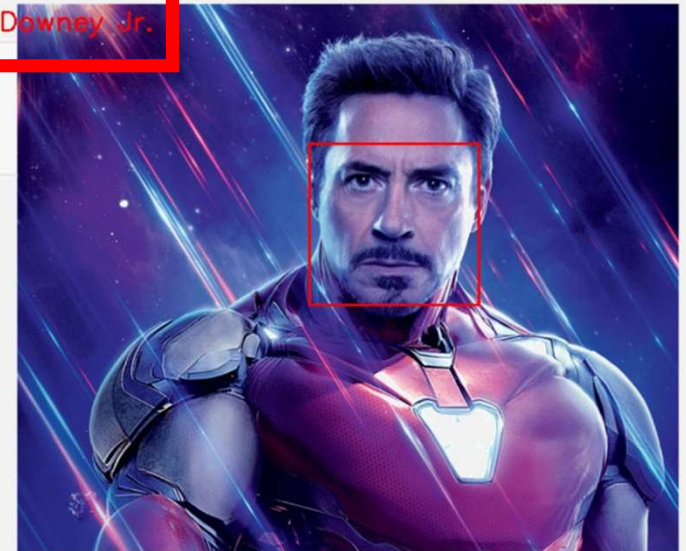
The Avengers

```
In [12]: matching_results = find_target_face()  
         recommend_matching_results(matching_results)
```

Recommended movie For you are:

The Avengers
Iron Man
Zodiac

Robert Downey Jr.



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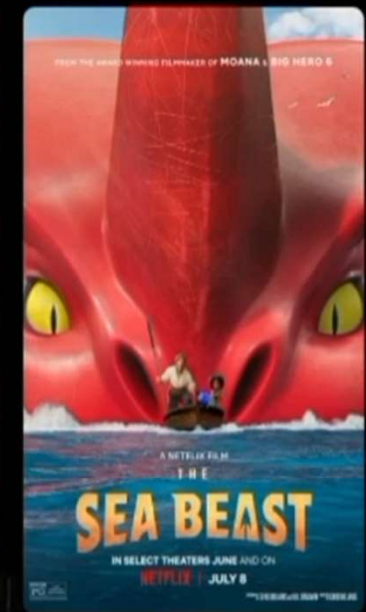
Discover Endless Streaming Bliss!

Entertainment on the Go, Wherever You Roam!

Ready to watch? Enter your email to embark on a thrilling journey through movies, TV shows, and so much more.

[Get Started >](#)

UPCOMING MOVIES



PLAUDED TV SHOWS



1. Developed IMDb movie review engine using Big Data technologies.
2. Conducted comprehensive analysis of movie reviews and metadata using Apache Pig.
3. Stored results in HDFS for scalable storage.
4. Implemented sentiment analysis to accurately predict user sentiment.
5. Provided personalized recommendations based on sentiment analysis results.
6. Integrated machine learning algorithms with Flask for a user-friendly interface.





Formats of Data

Utilizing different formats of data including semi-structured and unstructured to get more insights and information for the system.



Emerging Genres

Anticipate the rise of innovative film genres and storytelling styles that will captivate future audiences.



Shifting Audience Preferences

Can work on analyzing the results of recommendation system and review engine to predict the audience preferences from across the globe.



Real-Time Streaming Data Analysis

Can work on streaming data using Apache Spark to dynamically process data in run time.



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THANK YOU!!!



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